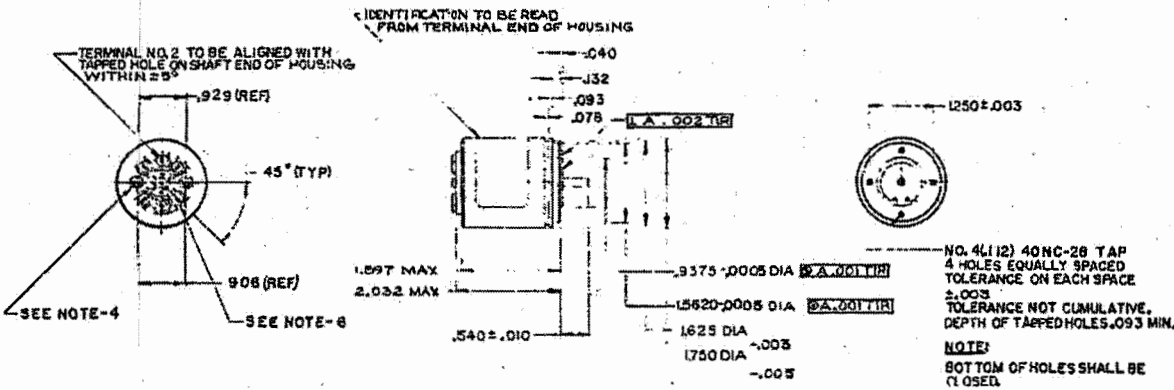
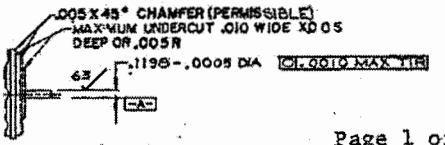
MILITARY SPECIFICATION SHEET

SERVOMOTOR, TYPE 18SM4 (7631932)

The complete requirements for procuring the servomotor described herein shall consist of this document and the issue in effect of Specification. MIL-S-22432.





FSC 6105
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NOTES:

1. Dimensions are in inches.

2. Unless otherwise specified, tolerances on decimals are .XXX ± .005 and angles ± 2°.

3. Perpendicularity and concentricity shall be measured with the unit in a vertical position supported by the shaft, the housing rotated.

4. Terminal sequence shall be shown and identified by molded lettering on the end cap or permanently affixed thereto. The end cap and terminal block shall be secured by screws other than the terminal screws. Mounting screw positions optional. Terminal block designed for MS17182 terminal lug.

5. Finish of the housing shall be bright metallic (passivate).

6. The following list of hardware shall be packaged in a ventilated envelope made of material having a PH value of 6-8 and compatible with the weight of the contents. The packaged hardware shall be placed in the unit servomotor package as follows:

Machine screw ---------- (MS 35276-213) 6 Required
Lockwasher ---------- (MS 35338-135) 6 Required

TABLE I
SERVOMOTOR, TYPE 18SM4 (7631932)

Performance Requirements and Characteristics

Requirements	Value	Unit	Tolerance
Electrical:			
Frequency:			
Rated	400	Hz	
Test	400	Hz	±1%
Range	380 to 420	Hz	
Voltage:			
Reference phase, rated	115	volts rms	
Reference phase, test	115	volts rms	±1%
Reference phase, range	103.5 to 126.5	volts rms	
Control phase, rated (series)	115	volts rms	
Control phase, test (series)	115	volts rms	±1%
Control phase, range (series)	0 to 126.5	volts rms	
Control phase, rated (parallel)	57.5	volts rms	
Control phase, range (parallel)	0 to 63.25	volts rms	

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TABLE I (Continued)

Requirements	Value	Unit	Tolerance
Current:			
Reference phase	.200	amperes rms	maximum
Control phase (series)	.200	amperes rms	maximum
Control phase (parallel)	.360	amperes rms	nominal
Power:			
Reference phase	9.9	watts	maximum
Control phase (series)	9.9	watts	maximum
Control phase (parallel)	8.5	watts	nominal
Synchronous Speed:			
Impedance:			
Reference phase	280+jx575	ohms	nominal
Control phase (series)	280+jx575	ohms	nominal
Power Factor:			
Reference phase	.43		nominal
Control phase (series)	.43		nominal
Effective Resistance:			
Control phase (series)	1460	ohms	nominal
Control phase (parallel)	365	ohms	nominal
Minimum Starting Voltage	3	volts rms	maximum
Coupling in Control Winding	.7	ratio	minimum
Mechanical:			
No-load Speed	4800	rpm	minimum
Stall Torque:			
at 23°C	2.25	ounce-inches	minimum
Rotor Moment of Inertia	4.0	$gm\text{-}cm^2$	maximum
Radial Play	.0008	inch	maximum
End Play	.0045	inch	maximum
	.0010	inch	minimum
Temperature Rise	85	degrees C	maximum
Noise, Mechanical	not excessive		
Time Constant	.0119	seconds	nominal
Reversing Time	.0201	seconds	nominal
Theoretical Acceleration at Stall	39,700	$radians/second^2$	nominal
Maximum Power Output	2.40	watts	nominal
Weight	14.5	ounces	maximum

NOTES:

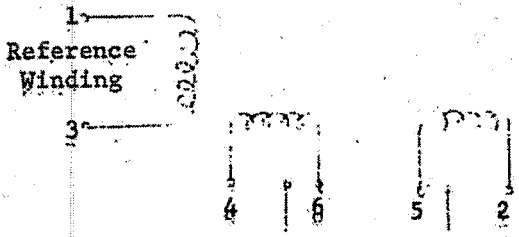
1. Type 18SM4 (7631932) servomotors shall be capable of operation in ambient temperatures ranging from -55°C to 100°C.

2. The servomotor identified herein shall conform to MIL-S-22432 with the following exceptions:

Salt Atmosphere
Audible Noise
Preproduction shall apply in lieu of qualification testing

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4. Connection Diagram:



Control Winding Excitation 400 cps
Figure 2

a. For CCW rotation the voltage at terminal 2 leads the voltage at terminal 1 by 90 electrical degrees.

b. For series operation of control winding terminals 5 and 6 are strapped and 115 volts rated is applied across terminals 2 and 4.

c. For multiple or parallel operation of control winding terminals 6 and 2 are strapped, and terminals 4 and 5 are strapped. 57.5 volts rated is applied across terminals 2 and 4.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 6105-A064

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